

Image Processing

INT3404 20

Week 11: Geometric transformation

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Schedule

Tuần	Nội dung	Yêu cầu đối với sinh viên (ngoài việc đọc tài liệu tham khảo)
1	Giới thiệu môn học	Cài đặt môi trường: Python 3, OpenCV 3, Numpy, Jupyter Notebook
2	Ảnh số (Digital image) – Phép toán điểm (Point operations) Làm quen với OpenCV + Python Điều chỉnh độ tương phản (Contrast adjust)– Ghép ảnh (Combining images)	
3	Histogram - Histogram equalization	Làm bài tập 1: điều chỉnh gamma tìm contrast hợp lý
4	Phép lọc trong không gian điểm ảnh (linear processing filtering)	Thực hành ở nhà
5	Phép lọc trong không gian điểm ảnh (linear processing filtering) (cont.) Thực hành: Cách tìm filters	Thực hành ở nhà
6	Thực hành: Ứng dụng của histogram; Tìm ảnh mẫu (Template matching)	Bài tập mid-term
	Trích rút đặc trưng của ảnh Cạnh (Edge) và đường (Line) và texture	Thực hành ở nhà
8	Các phép biến đổi hình thái (Morphological operations)	Làm bài tập 2: tìm barcode
9	Chuyển đổi không gian – Miền tần số – Phép lọc trên miền tần số Thông báo liên quan đồ án môn học	Đăng ký thực hiện đồ án môn học
10	Xử lý ảnh màu (Color digital image processing)	Làm bài tập 3: Chuyển đổi mô hình màu và thực hiện phân vùng
11	Các phép biến đổi hình học (Geometric transformations)	Thực hành ở nhà
12	Nhiều – Mô hình nhiễu – Khôi phục ảnh (Noise and restoration)	Thực hành ở nhà
13	Nén ảnh (Compression)	Thực hành ở nhà
14	Hướng dẫn thực hiện đồ án môn học	Trình bày đồ án môn học
15	Hướng dẫn thực hiện đồ án môn học Tổng kết cuối kỳ	Trình bày đồ án môn học

Recall week 10: Color image processing

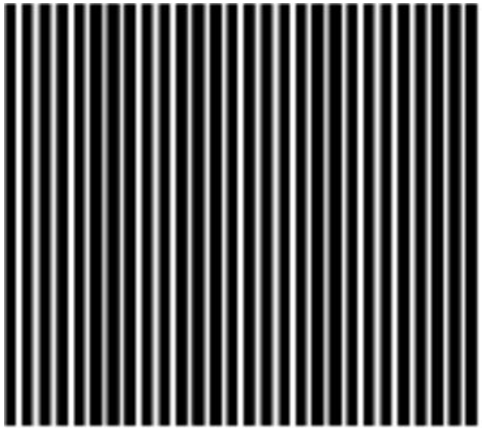
- Color phenomenon
- Color matching/creation: additive vs subtractive
- Attributes of colors:
 - Hue
 - Saturation, Chroma, Colorfulness
 - Brightness, Lightness, Value, Intensity
- Color space: RGB, HSV, CYMK, YCbCr
- Image segmentation using color information

Frequency: Highpass $\sim$ edge

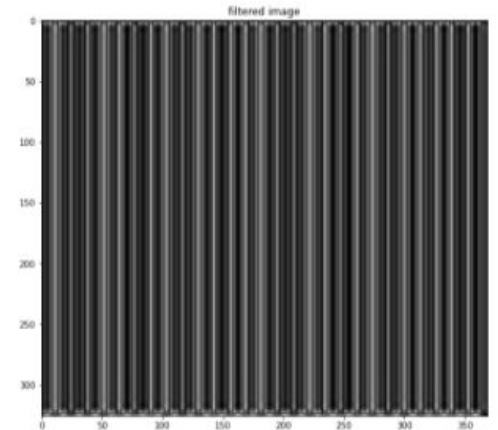
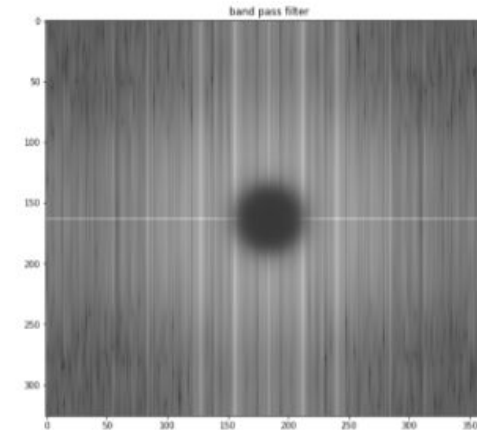
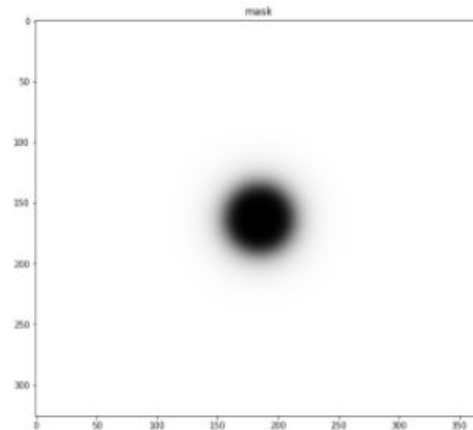
$$F[k, l] = \frac{1}{MN} \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f[m, n] e^{-j2\pi \left(\frac{k}{M}m + \frac{l}{N}n \right)}$$

- "Frequency":
 - Frequency of changing
 - Frequency of events

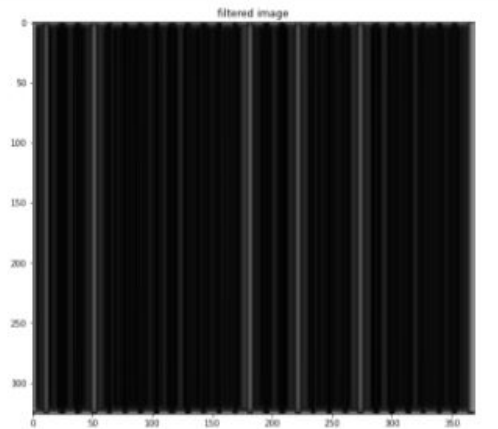
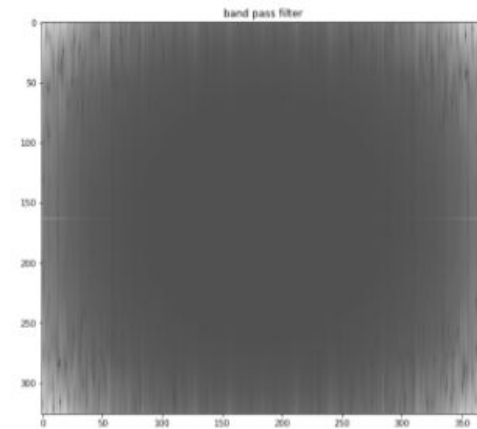
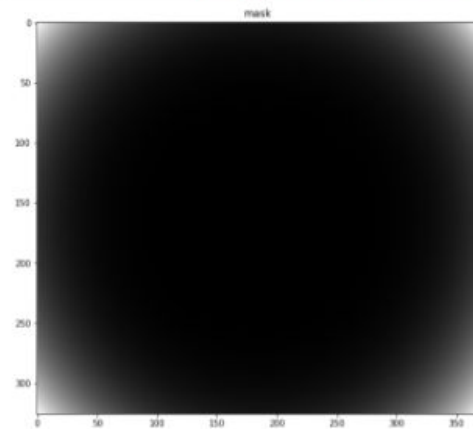
High pass



```
1 filter_img('highfreq.png', 30)
```

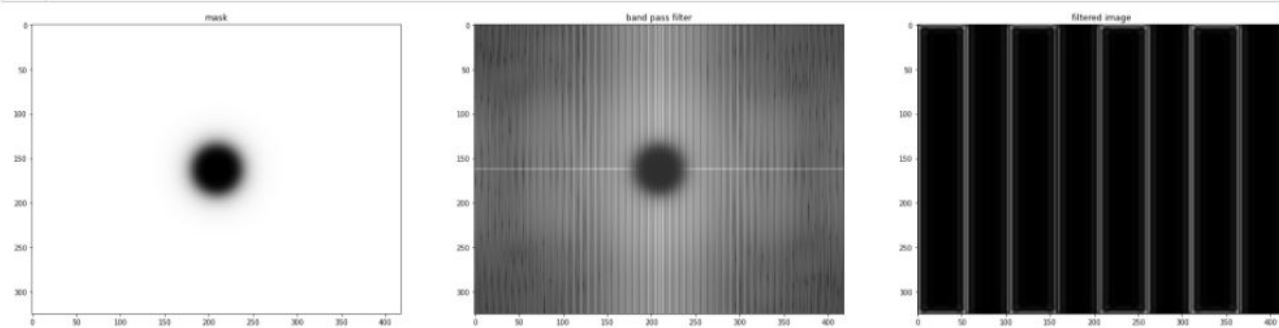


```
1 filter_img('highfreq.png', 400)
```

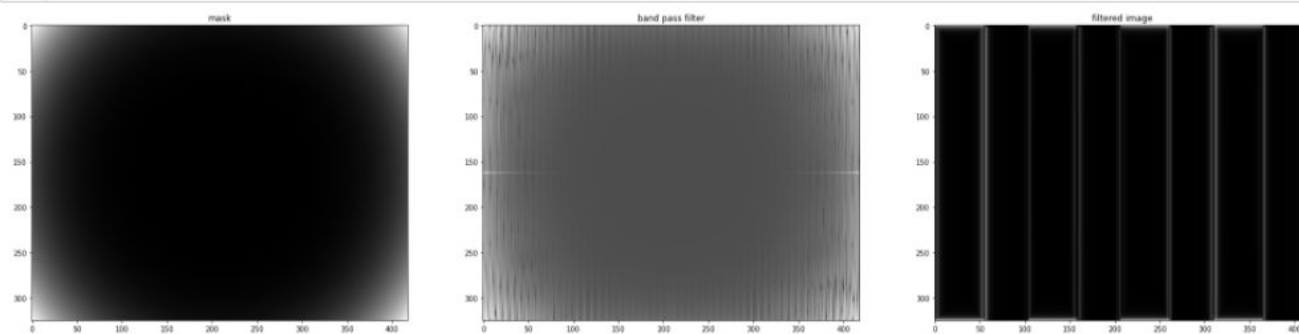




```
1 filter_img('lowfreq.png', 30)
```



```
1 filter_img('lowfreq.png', 400)
```



Geometric transformation

What is Geometric transformation?

- So far, image processing operations □ modify the color values of pixels in a given image
- Geometric transformation □ modify the positions of pixels in an image, but keep the color unchanged
- Purpose:
 - To create special effects
 - To register two images taken of the same scene at different times
 - To morph one image to another



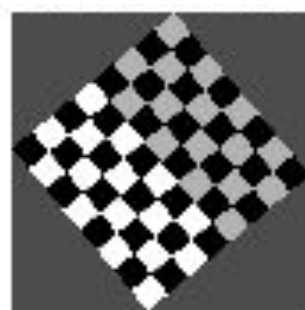
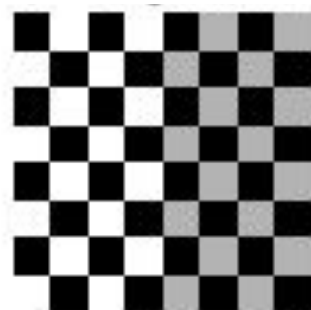
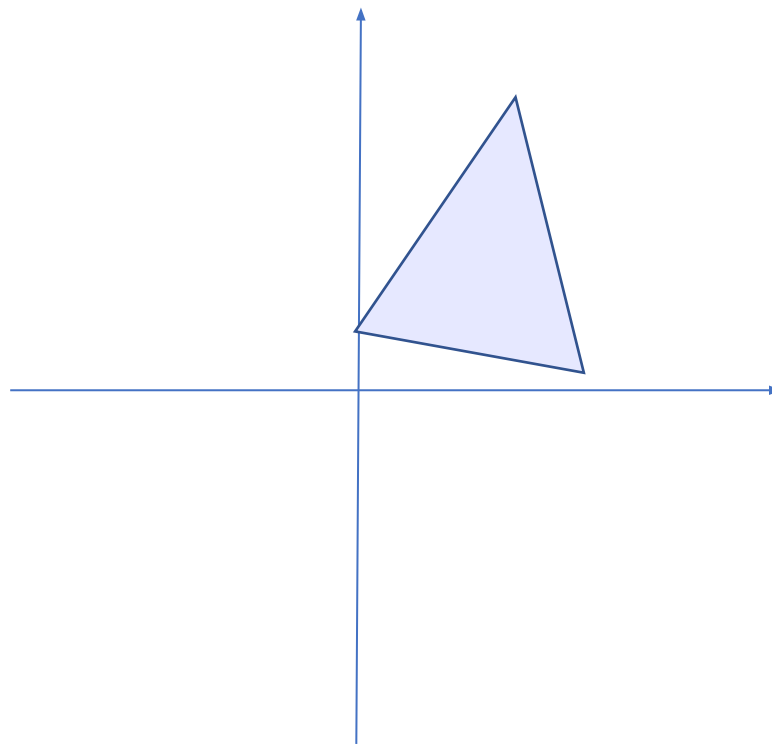
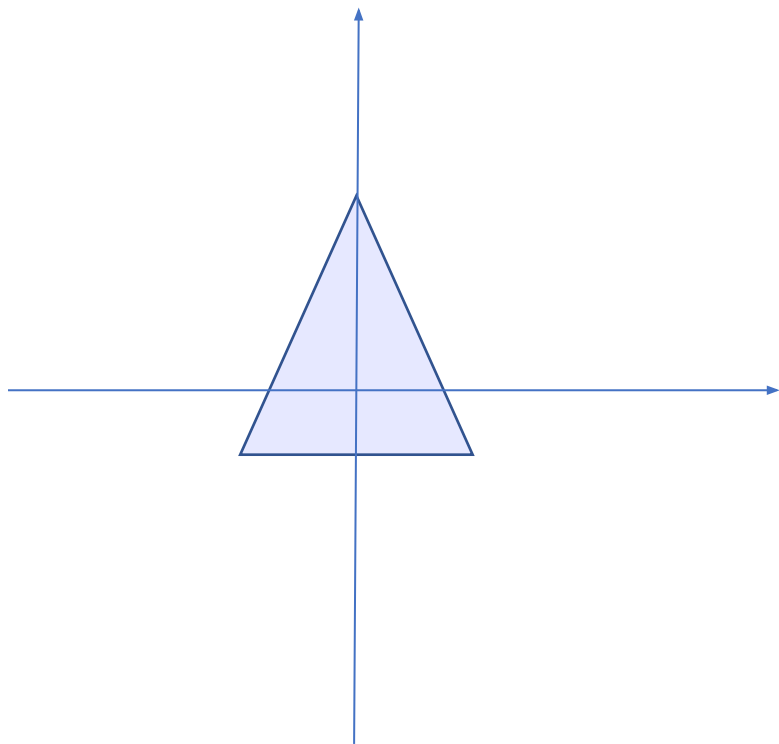
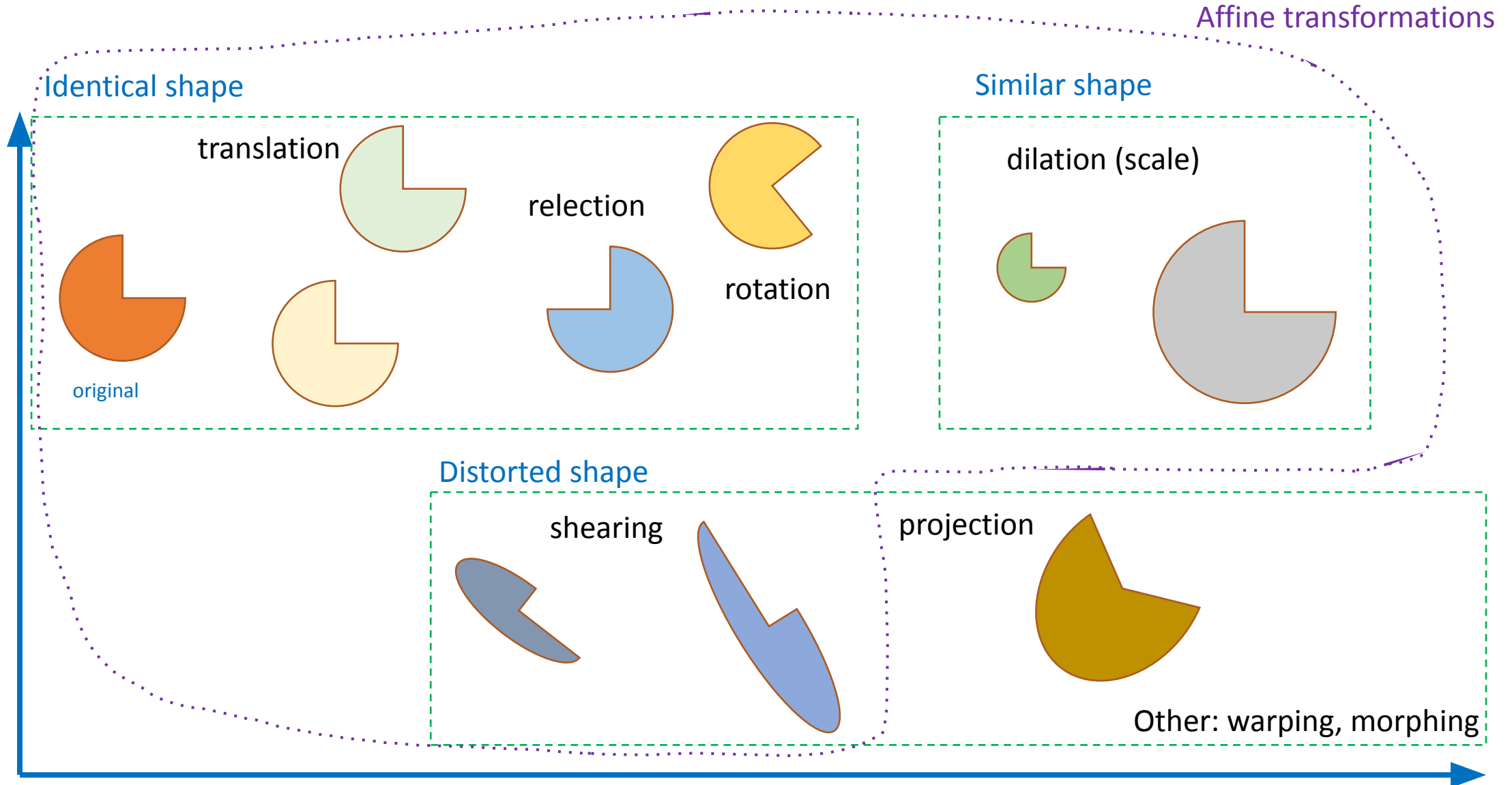


Illustration of mapping functions



Geometric transformations

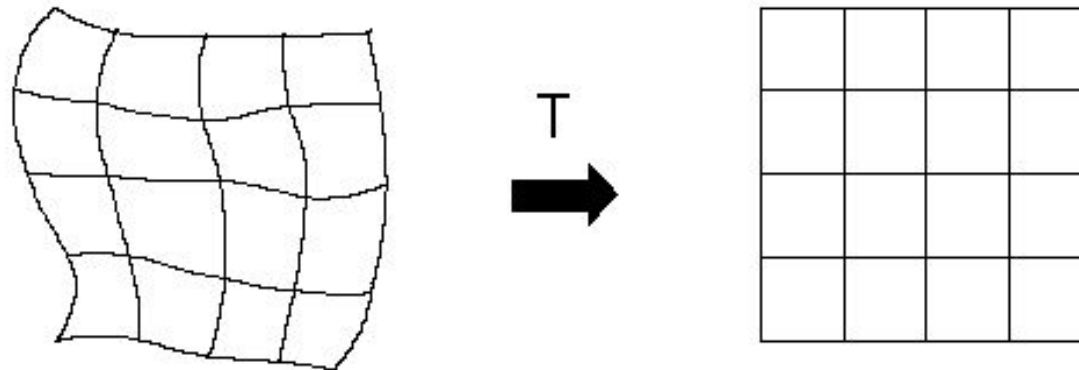


Two basic operations of geometric transformation

1. Spatial transformation of coordinates

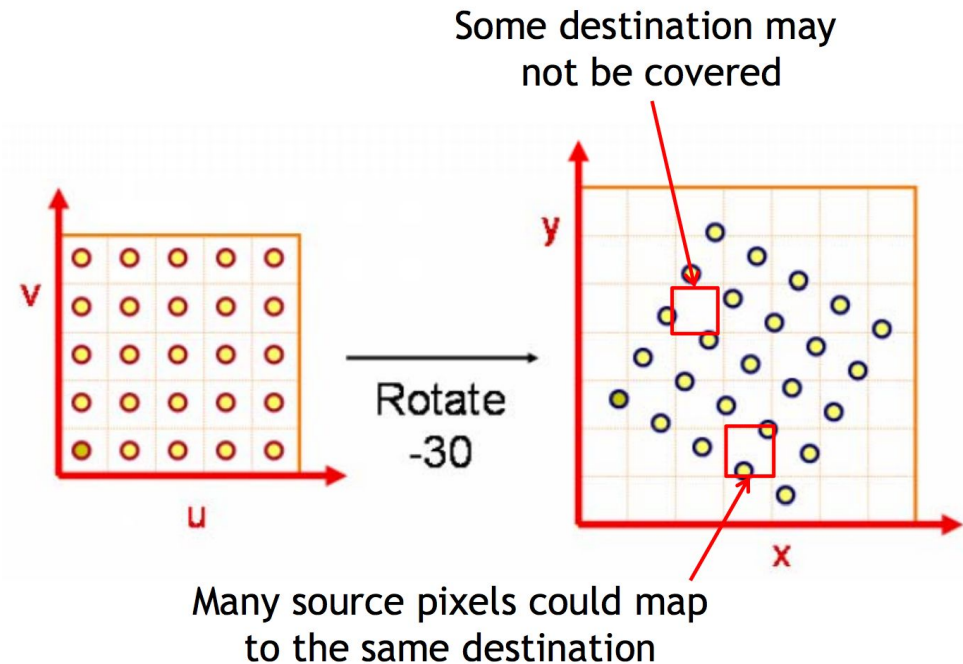
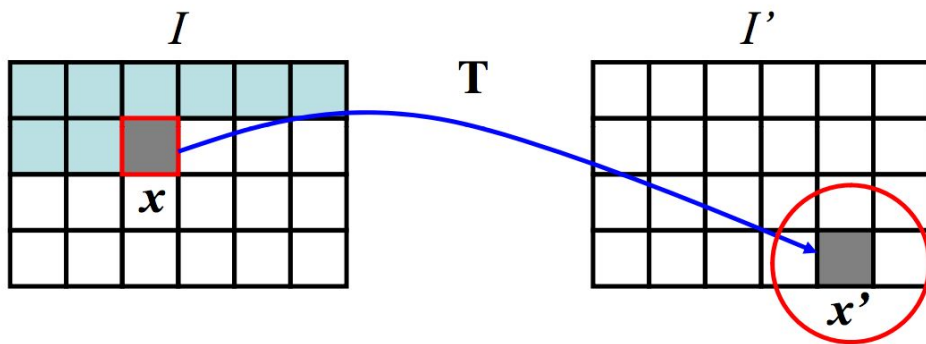
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \mathbf{T} \begin{bmatrix} x \\ y \end{bmatrix}$$

2. Intensity interpolation that assigns intensity values to the spatially transformed pixels



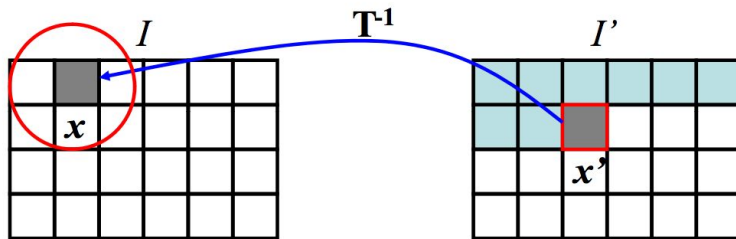
Forward vs backward (inverse) mapping

- Forward mapping: $\begin{bmatrix} x' \\ y' \end{bmatrix} = T \begin{bmatrix} x \\ y \end{bmatrix}$
 - Iterates over pixels of the input image (x, y) , computes new location (x', y') , and copies its values to the new location

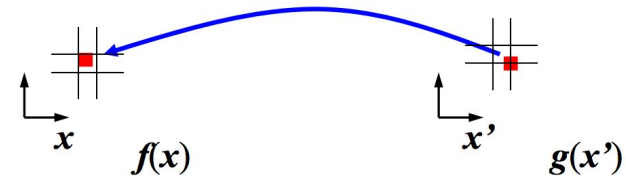


Forward vs backward (inverse) mapping

- Backward mapping: $\begin{bmatrix} x \\ y \end{bmatrix} = T^{-1} \begin{bmatrix} x' \\ y' \end{bmatrix}$
 - Iterates over pixels of the output image and uses the inverse transformation to determine the position in the input image from which a value must be sampled

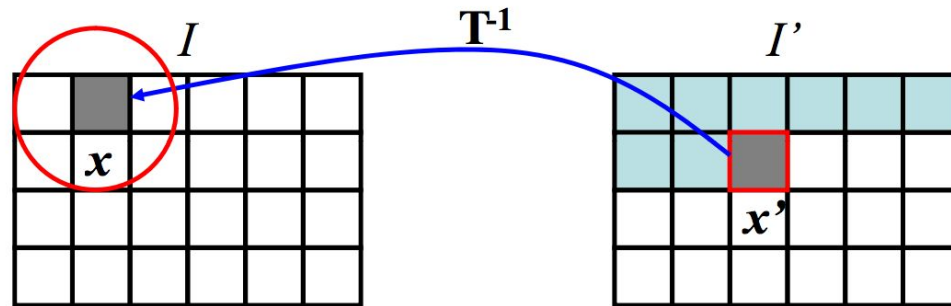


- What if pixel comes from “between” two pixels?
- Answer: *resample* color value from *interpolated (prefiltered)* source image



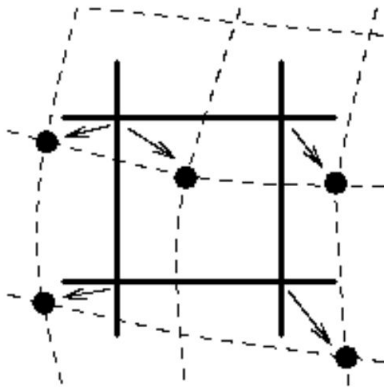
Inverse warping

```
iwarp(I, I', T)
{
  for (y=0; y<I'.height; y++)
    for (x=0; x<I'.width; x++) {
      (x,y)=T-1(x',y');
      I'(x',y')=Reconstruct(I,x,y,kernel);
    }
}
```



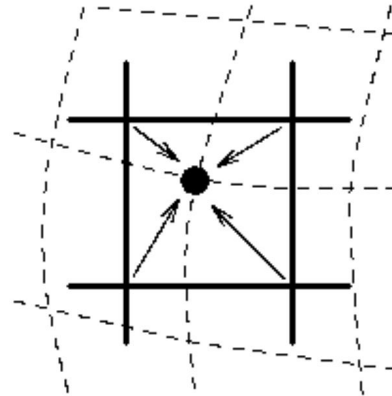
Intensity interpolation

Nearest neighbor



Assigns to the new location the intensity of its nearest neighbor

Bilinear

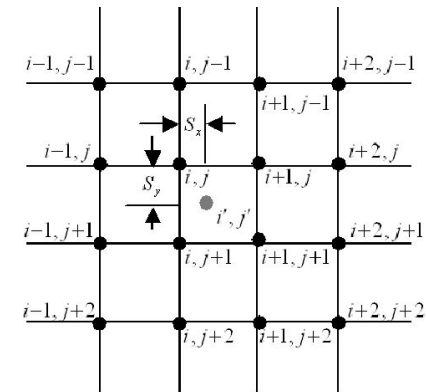


Use four nearest neighbors to estimate the intensity at a given location

$$v(x, y) = ax + by + cxy + d$$

-> solve for a, b, c, d

Bicubic

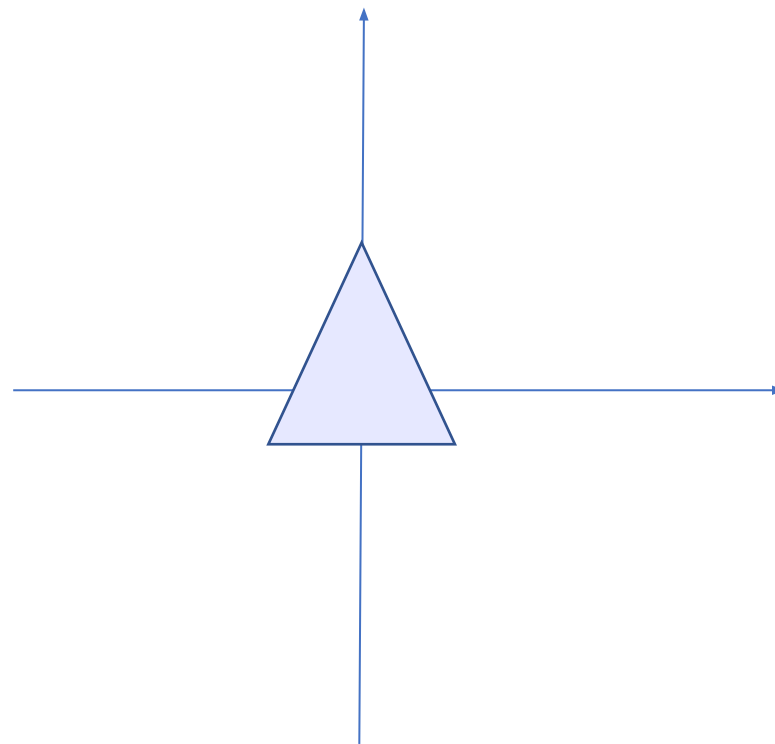
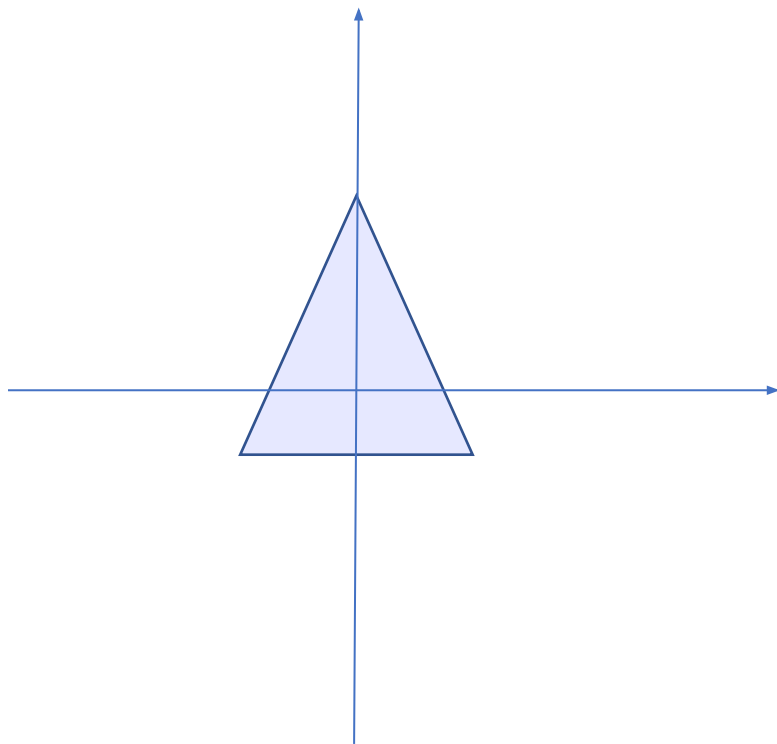


Use sixteen nearest neighbors to estimate the intensity at a given location

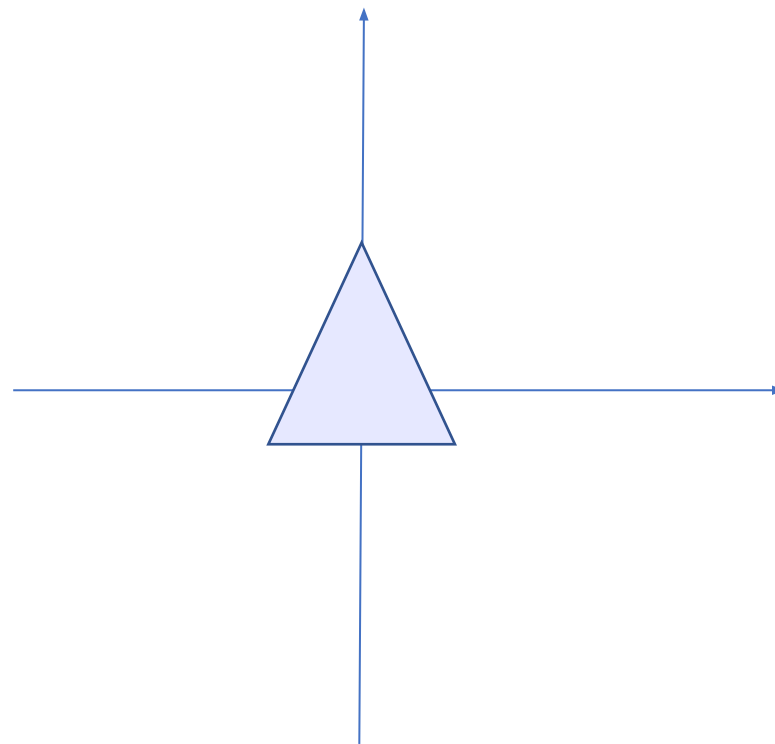
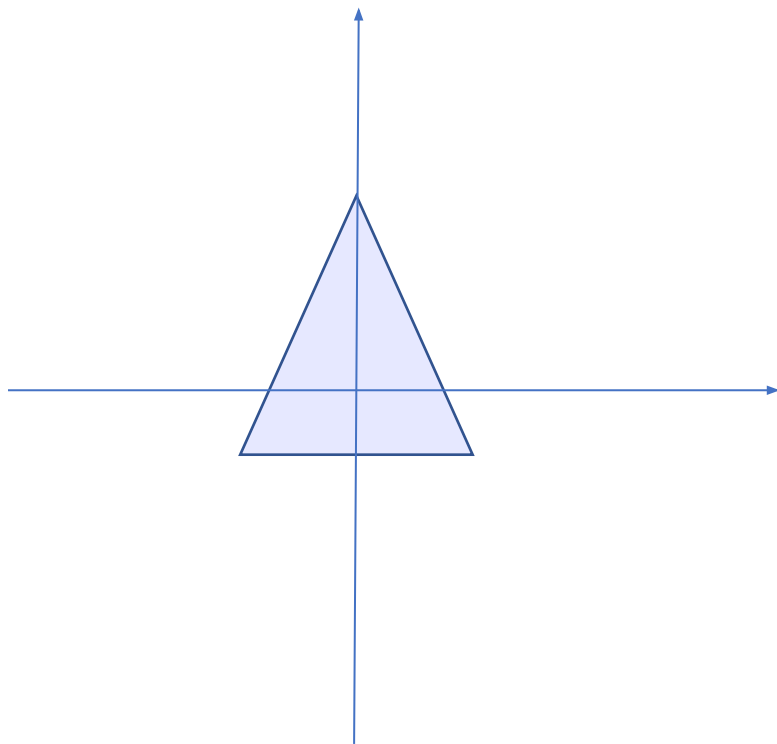
$$v(x, y) = \sum_{i=0}^3 \sum_{j=0}^3 a_{ij} x^i y^j$$

-> solve for a_{ij}

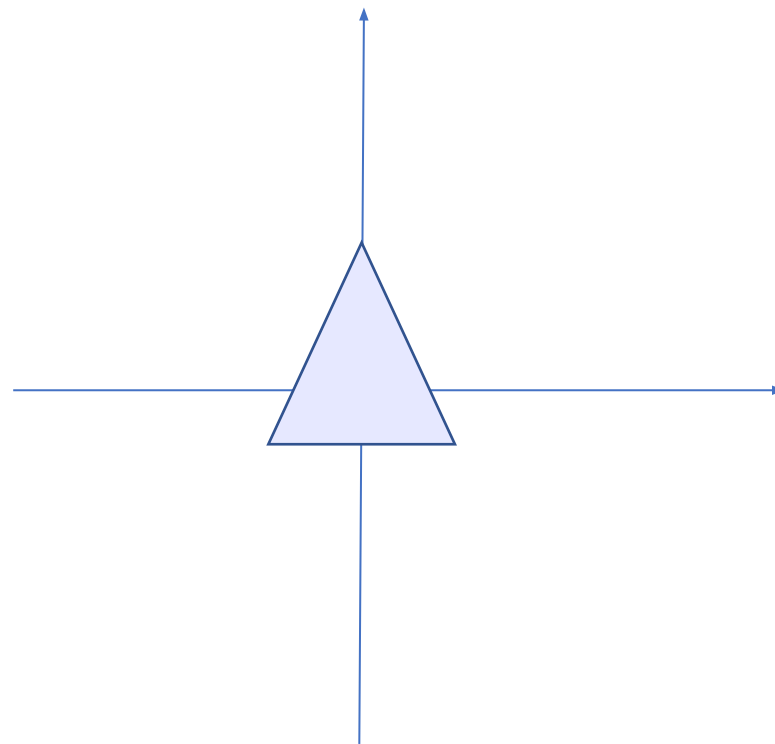
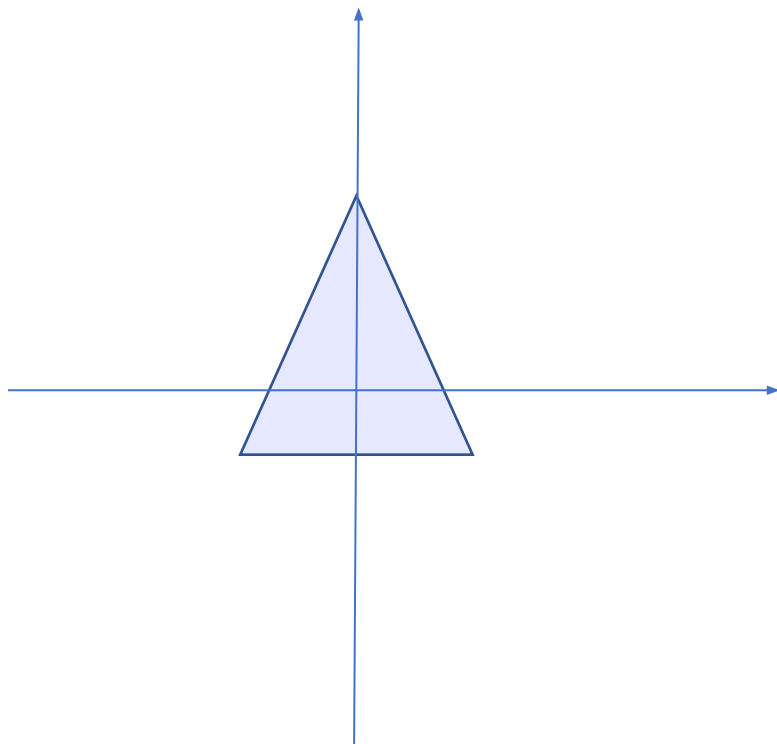
Scaling



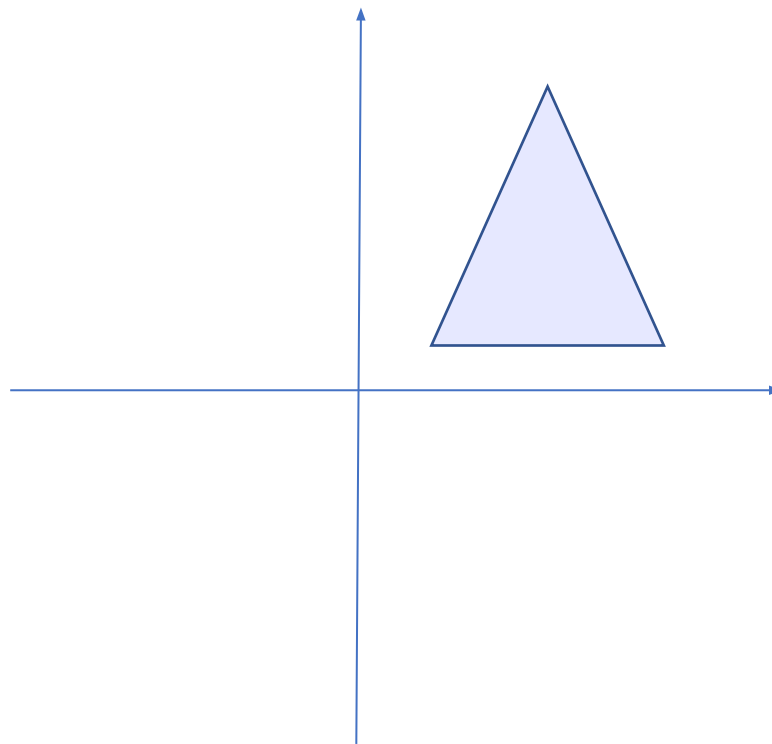
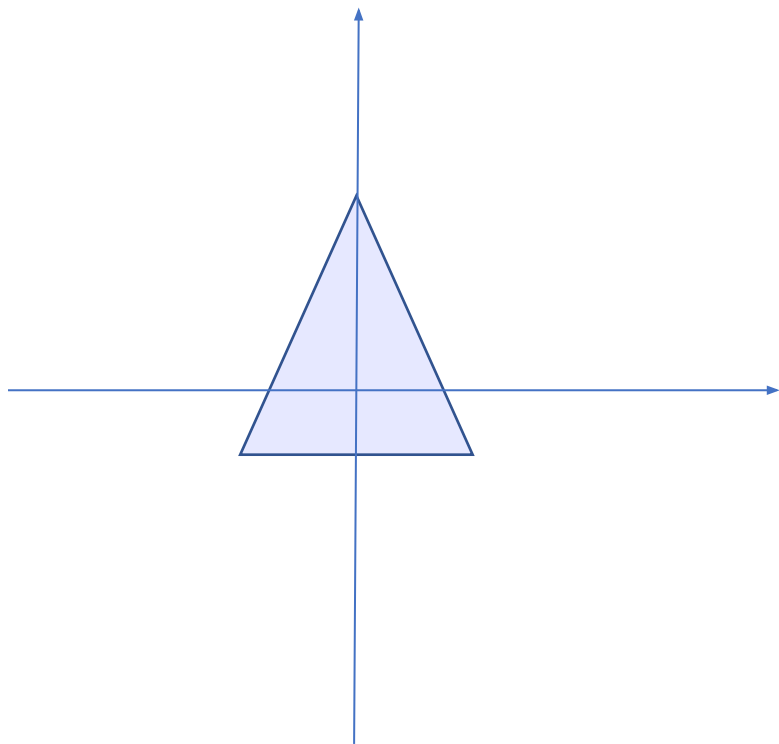
Scaling



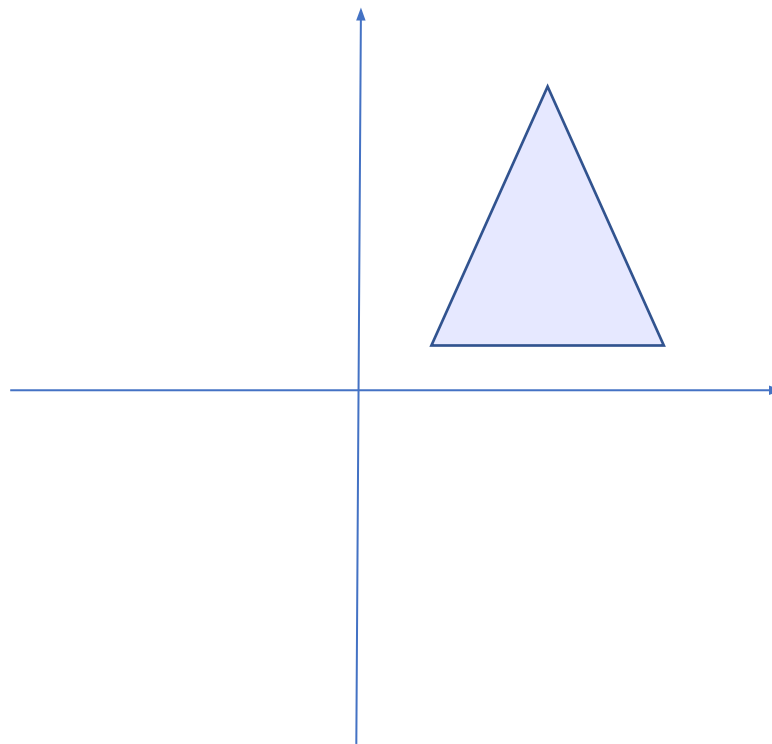
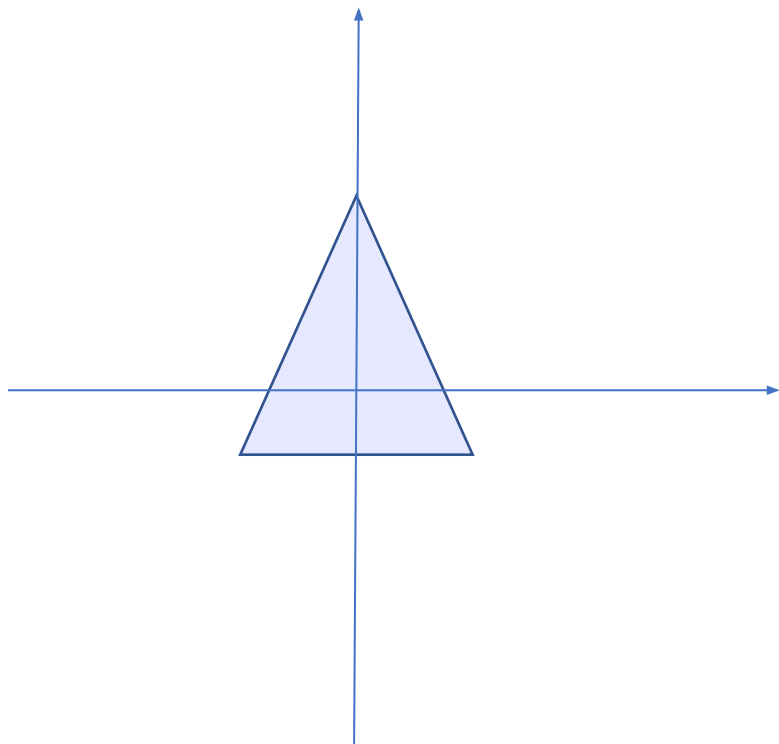
Scaling



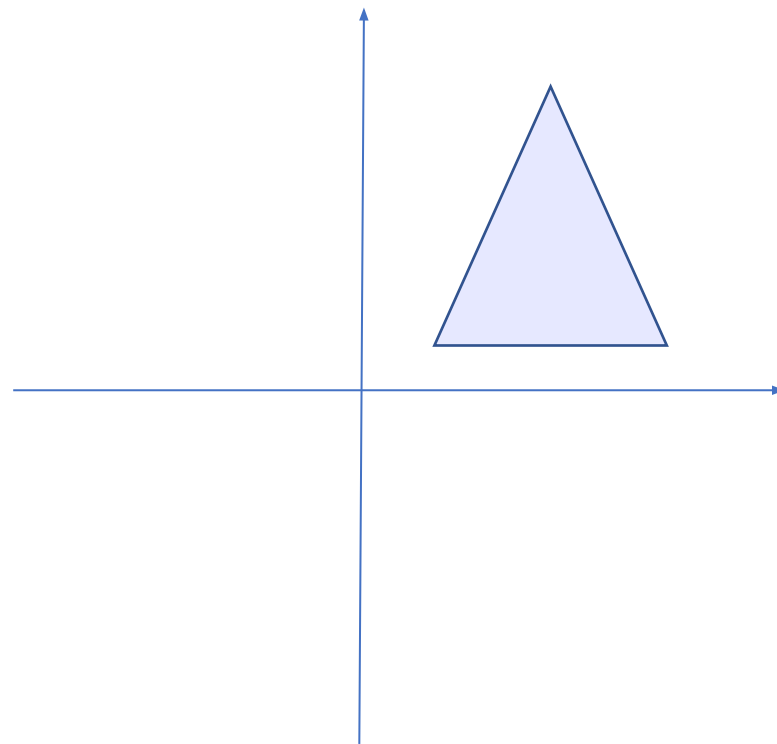
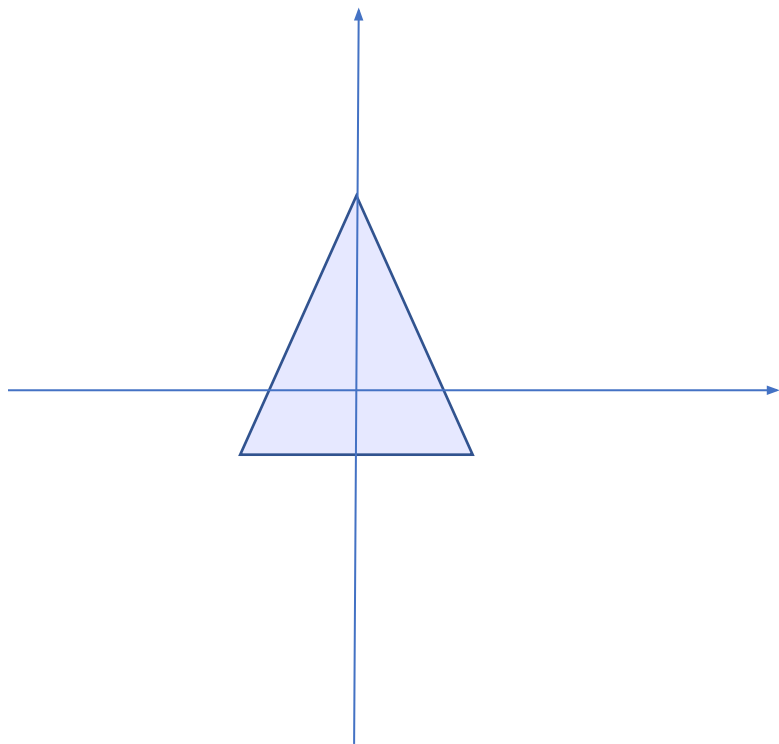
Translation



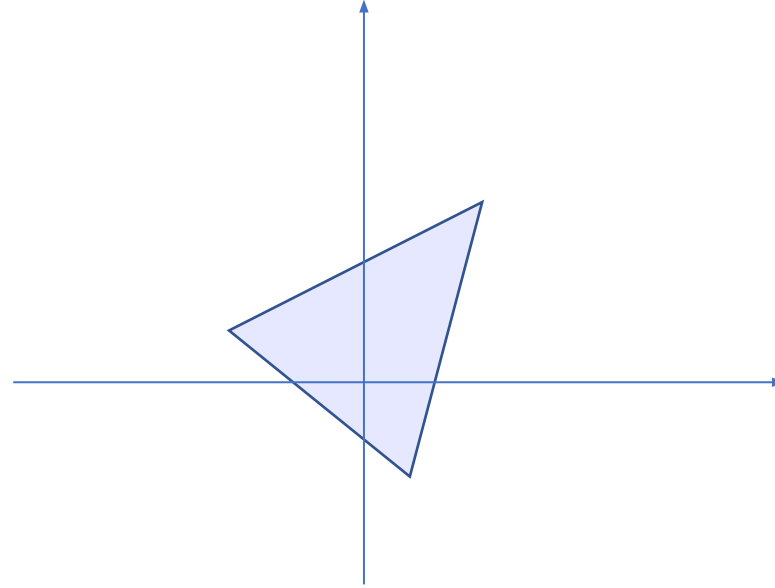
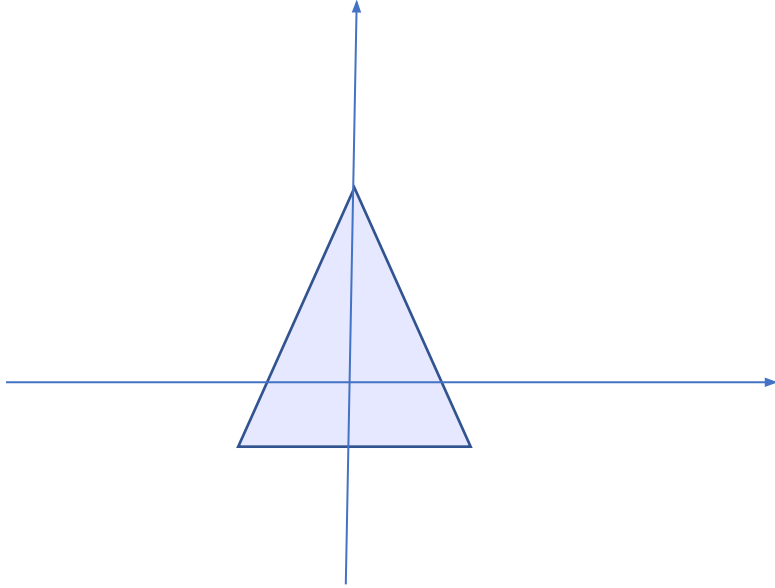
Translation

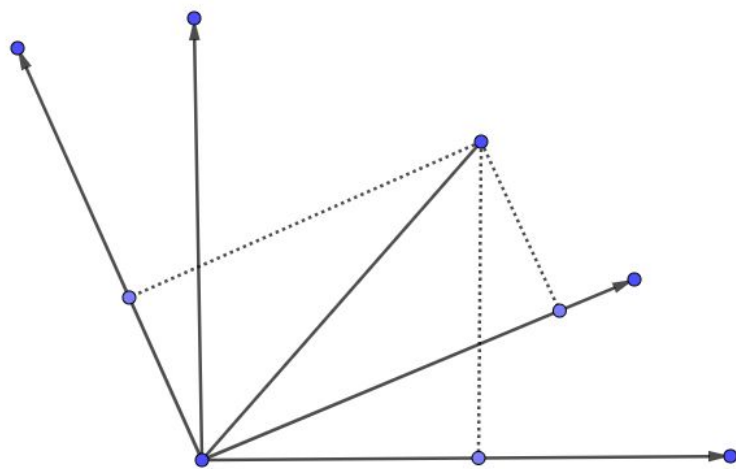


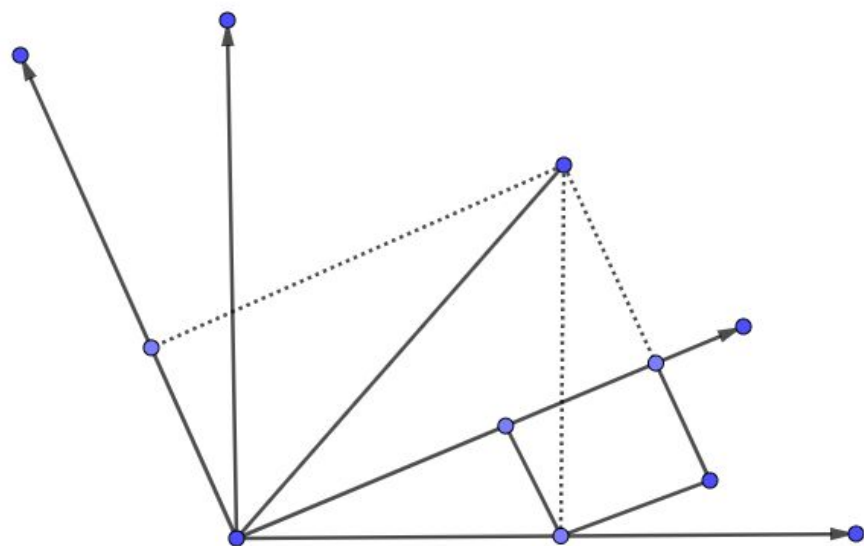
Translation



Rotation







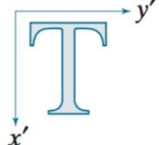
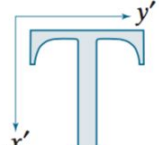
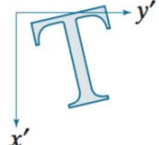
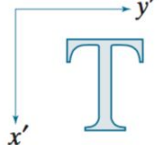
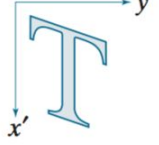
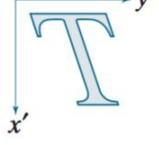
Affine transformation

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \mathbf{T} \begin{bmatrix} x \\ y \end{bmatrix}$$



Expressed using a 3x3 matrix

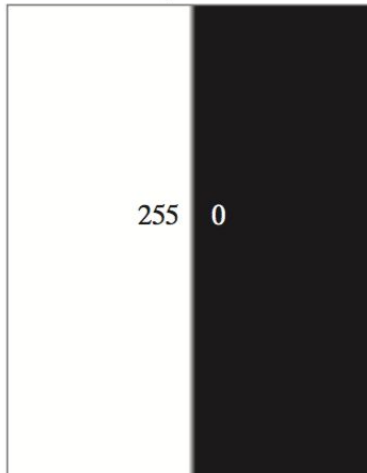
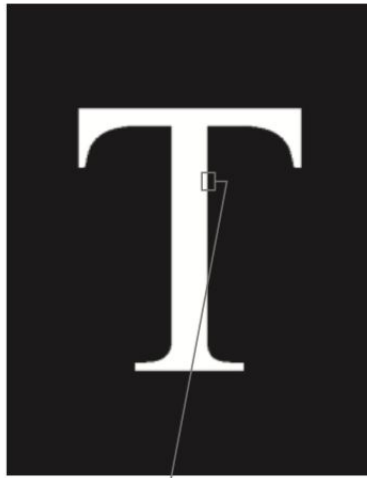
$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \mathbf{A} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Transformation Name	Affine Matrix, A	Coordinate Equations	Example
Identity	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x$ $y' = y$	
Scaling/Reflection (For reflection, set one scaling factor to -1 and the other to 0)	$\begin{bmatrix} c_x & 0 & 0 \\ 0 & c_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = c_x x$ $y' = c_y y$	
Rotation (about the origin)	$\begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x \cos \theta - y \sin \theta$ $y' = x \sin \theta + y \cos \theta$	
Translation	$\begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x + t_x$ $y' = y + t_y$	
Shear (vertical)	$\begin{bmatrix} 1 & s_v & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x + s_v y$ $y' = y$	
Shear (horizontal)	$\begin{bmatrix} 1 & 0 & 0 \\ s_h & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x' = x$ $y' = s_h x + y$	

Example of interpolation

Rotated -21°

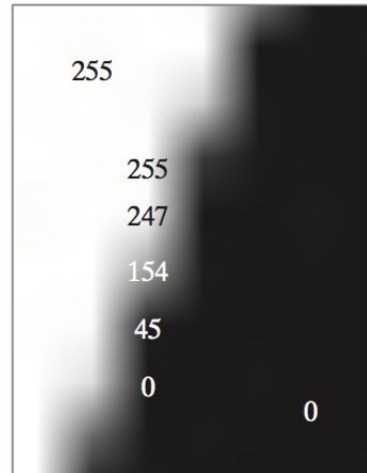
Nearest neighbor



Bilinear

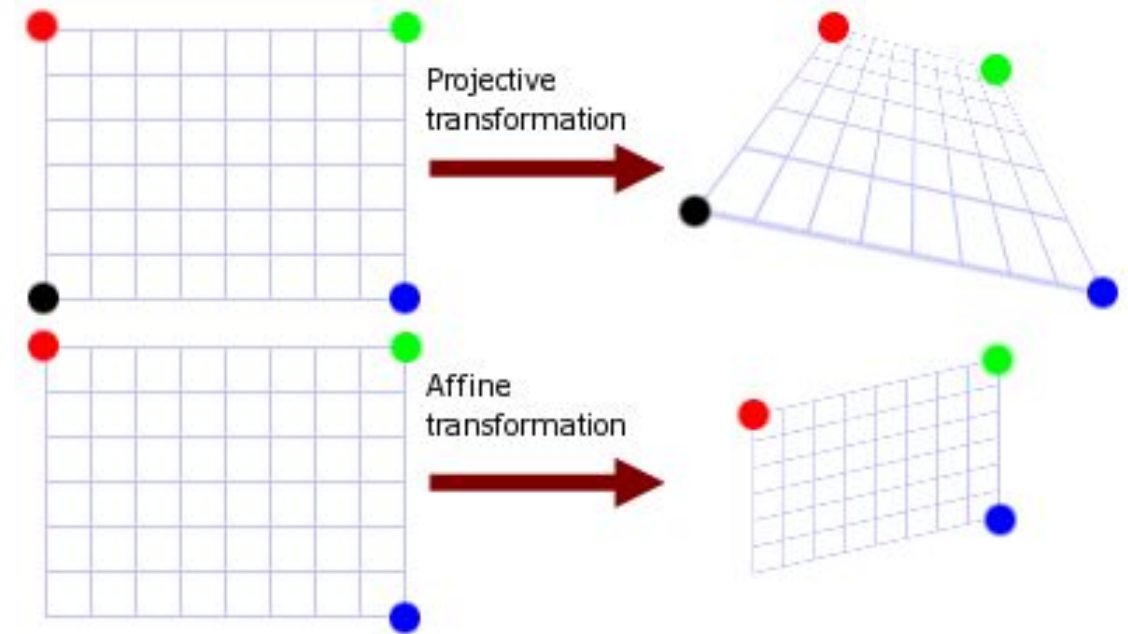


Bicubic



Linear transformation

- Two types of linear transformation:
 - Affine transformation:
 - Preserve parallelism, length, and angle
 - Projective transformation:
 - Preserve collinearity, and incidence



General transformation matrix

Translation vector

Rotation matrix

$$\begin{bmatrix} a1 & a2 & b1 \\ a3 & a4 & b2 \\ c1 & c2 & 1 \end{bmatrix}$$

Projection vector

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} \approx \begin{bmatrix} wx' \\ wy' \\ w \end{bmatrix} = \begin{bmatrix} a1 & a2 & b1 \\ a3 & a4 & b2 \\ c1 & c2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

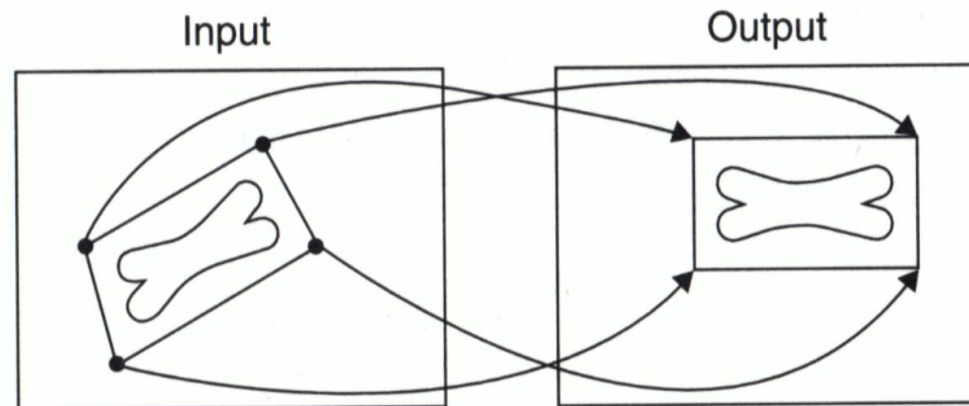
Also referred to as “homography matrix”

General transformations

- Control points
- Projective transformation
- Image warp and image morphing

General transformations

- Till now: rotation, scaling, translation (shift), projective transformation are all linear in x and y
- More general transformations: polynormal transformations
 - Order is higher than 1
- Extract transformation is derived from control points
 - a point in the input image and its corresponding point in the output image



Polynomial transformation

- Assume $a(x,y)$ and $b(x, y)$ as polynomials of order N with unknown coefficients

$$x' = a(x,y) = \sum_{i=0}^N \sum_{j=0}^{N-i} a_{ij} x^i y^j$$

$$y' = b(x,y) = \sum_{i=0}^N \sum_{j=0}^{N-i} b_{ij} x^i y^j$$

Determining the coefficients requires at least as many control points as the polynomials have coefficients.

Polynomial transformation with order N=2

$$\begin{aligned}x' &= a(x,y) = a_{00} + a_{10}x + a_{01}y + a_{11}xy + a_{20}x^2 + a_{02}y^2 \\y' &= b(x,y) = b_{00} + b_{10}x + b_{01}y + b_{11}xy + b_{20}x^2 + b_{02}y^2\end{aligned}$$

a_{00}, b_{00} : Shift vector

a_{10}, b_{01} : Linear scaling in x, y direction

a_{01}, b_{10} : Shear in x, y direction¹

a_{11}, b_{11} : y -dependent scale in x , x -dependent scale in y

a_{20}, b_{02} : non-linear (quadratic) scale in x, y

¹A rotation can be described as a combination of shear and linear scaling first in one, then the other coordinate: Any Rotation by angle $\theta \neq \pm 90^\circ$ can be decomposed in the following way:

$$\begin{bmatrix} a_{10} & a_{01} \\ b_{10} & b_{01} \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} = \begin{bmatrix} 1/\cos \theta & \sin \theta / \cos \theta \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -\sin \theta & \cos \theta \end{bmatrix} \quad (4.18)$$

The first (the rightmost one) is a 1D scale and shear in y , the second (the left one) is a 1D scale and shear in x .

Example of polynomial geometric warps

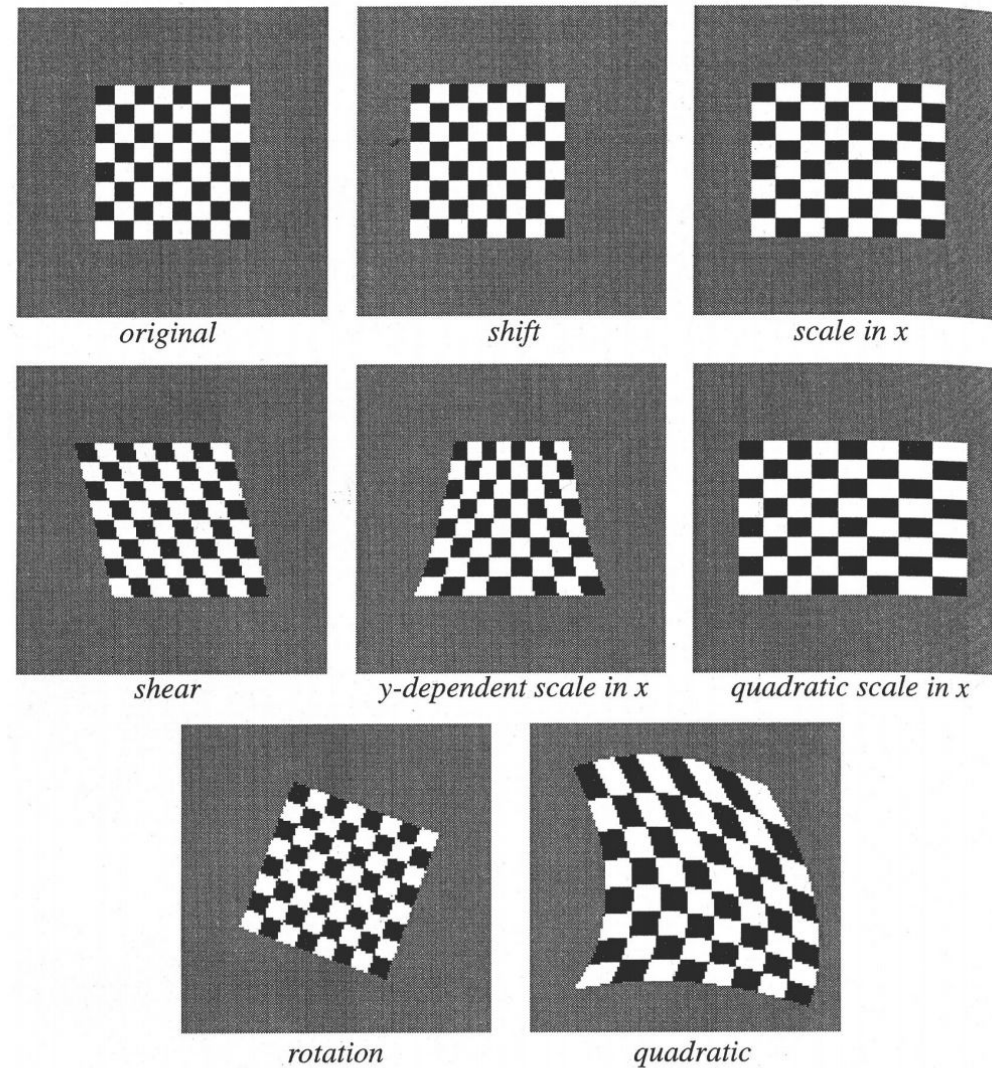


Fig. 4.9: Some polynomial geometric warps (Fig 7-30 in Schowengerdt, 1997)

Applications of geometric transformation

- Geometric calibration/Image rectification
 - Remove camera-induced distortion, i.e., convert non-rectangular pixel coordinates to rectangular coordinates
- Image registration
 - Geometrically match two images or an image and a map; stationary objects should have same position in both images
- Map projections

Image registration example

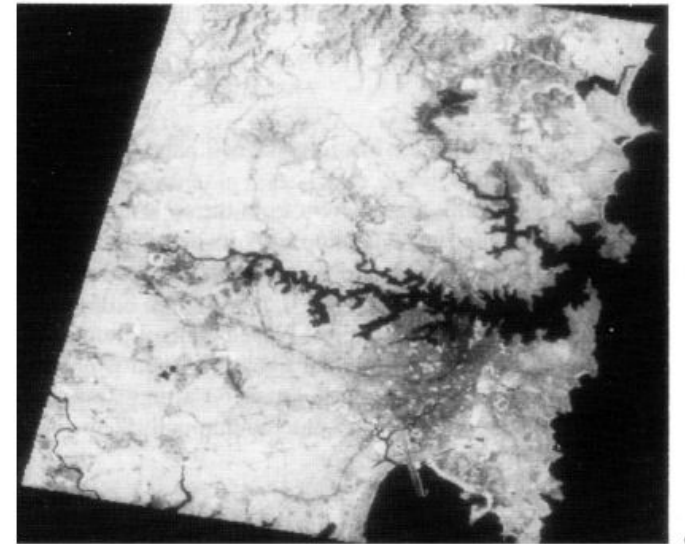
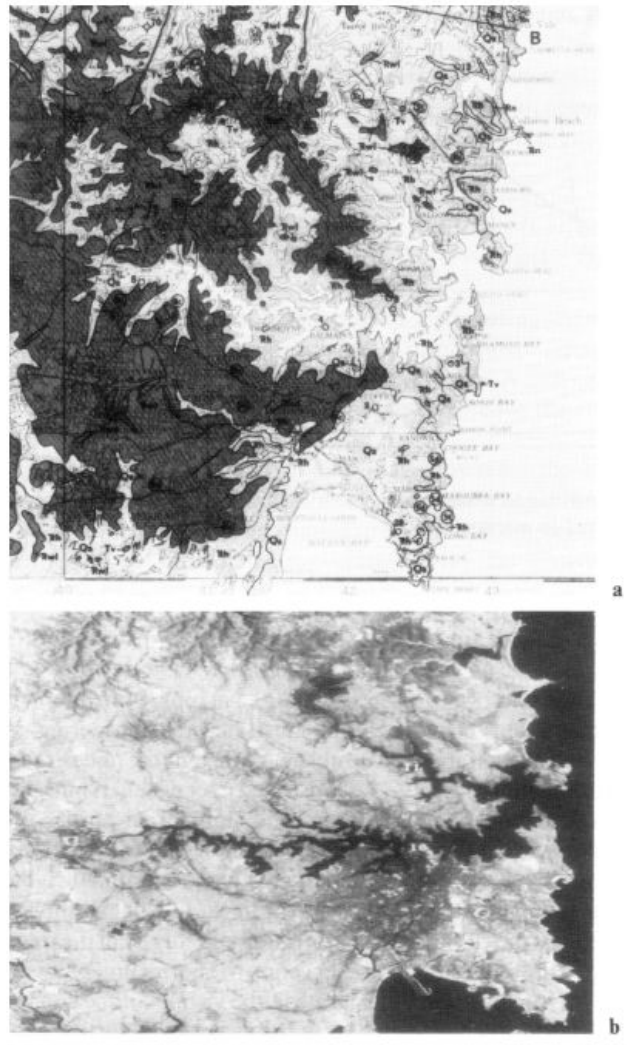


Fig. 4.10: Image registration. (a) Map; (b) Landsat MSS image to be registered; (c) Landsat image registered to map using 2nd order polynomials (Fig. 2.16 from Richards, 1986)

Example of image rectification

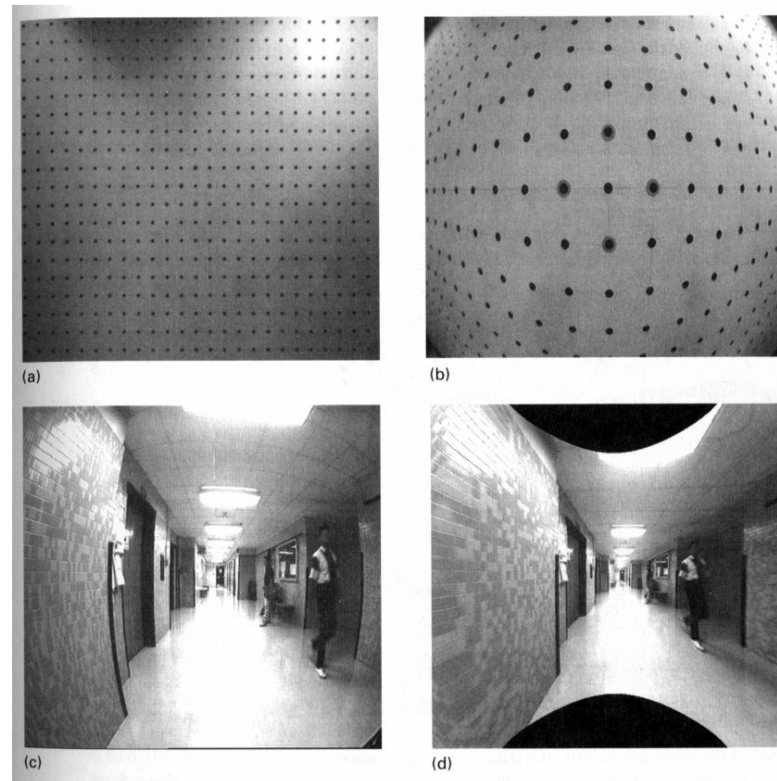


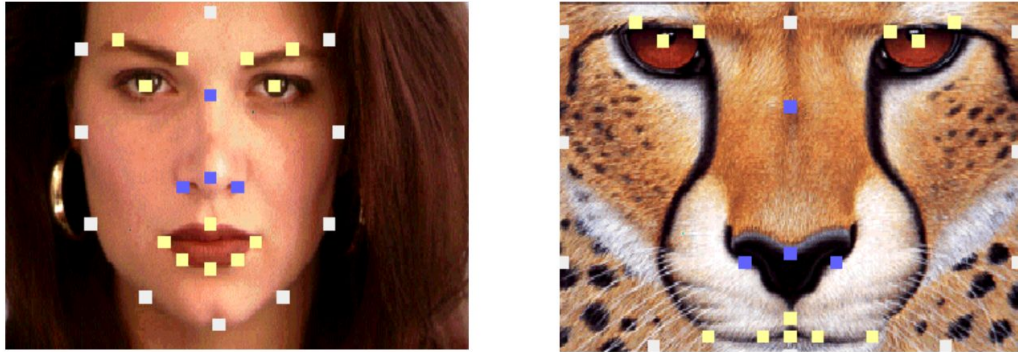
Fig. 4.8: Geometric rectification of an image taken with a fish-eye lens: (a) test target, (b) fish-eye image of test target, (c) fish-eye image (d) rectified image (Fig 8.9 from Castleman, 1996)

References

- Gonzalez book
- (Chap 4) Geometric transformations
 - <http://www.iup.uni-bremen.de/~melsheim/dip/dipSS17-L3.pdf>
- Image warping/morphing
 - https://www.csie.ntu.edu.tw/~cyy/courses/vfx/18spring/lectures/handouts/lec05_morphing.pdf

Mesh warping

Specify corresponding points



Convert to mesh warping



Interpolate to a complete warping function



warp

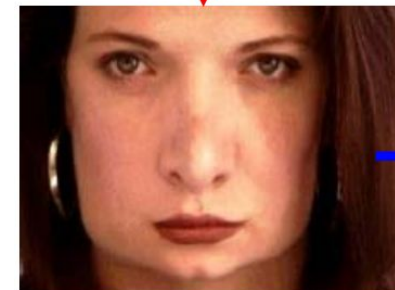


Image morphing

- To synthesize a fluid transformation from one image to another

